

120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17

Day 117: Monte Carlo Greeks for Risk Management

1 Introduction

In practice, many derivatives do not have closed-form pricing formulas. As a result, sensitivities (Greeks) must be computed numerically.

Monte Carlo methods provide a flexible framework for estimating both prices and Greeks.

2 Greeks Definition

For a derivative price $V(S, \sigma, t)$:

- Delta: $\Delta = \frac{\partial V}{\partial S}$
- Gamma: $\Gamma = \frac{\partial^2 V}{\partial S^2}$
- Vega: $\nu = \frac{\partial V}{\partial \sigma}$

3 Monte Carlo Pricing

Under risk-neutral measure:

$$V = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{Payoff}(S_T)]$$

We approximate:

$$V \approx e^{-rT} \frac{1}{N} \sum_{i=1}^N \text{Payoff}(S_T^{(i)})$$

4 Finite Difference Method

Greeks can be estimated by perturbing inputs:

4.1 Delta

$$\Delta \approx \frac{V(S+h) - V(S-h)}{2h}$$

4.2 Gamma

$$\Gamma \approx \frac{V(S+h) - 2V(S) + V(S-h)}{h^2}$$

4.3 Vega

$$\nu \approx \frac{V(\sigma+h) - V(\sigma-h)}{2h}$$

5 Common Random Numbers (CRN)

To reduce variance, use the same random numbers for perturbed simulations.

This improves numerical stability.

6 Algorithm

1. Simulate asset paths under baseline parameters
2. Perturb input parameter (e.g., S , σ)
3. Recompute option price
4. Apply finite difference formulas

7 Variance Reduction

- common random numbers
- antithetic variates
- control variates

8 Applications

- hedging strategies
- risk management
- exotic option pricing

9 Limitations

- computationally expensive
- sensitive to step size h

10 Conclusion

Monte Carlo Greeks provide a flexible and practical approach to risk estimation, especially when analytical solutions are unavailable.